

Practical 9

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code:-

```
MINGW64/c/Users/STUDENTS/Desktop
GNU nano 8.7 sstf.c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, head, i, j, total_movement = 0;
    printf("Enter number of requests: ");
    scanf("%d", &n);
    int req[n], visited[n];
    printf("Enter the request sequence:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &req[i]);
        visited[i] = 0;
    }
    printf("Enter initial head position: ");
    scanf("%d", &head);
    for(i = 0; i < n; i++) {
        int min = 9999, index = -1;
        for(j = 0; j < n; j++) {
            if(!visited[j]) {
                int distance = abs(head - req[j]);
                if(distance < min) {
                    min = distance;
                    index = j;
                }
            }
        }
        total_movement += min;
        head = req[index];
        visited[index] = 1;
    }
    printf("\nTotal Head Movement = %d\n", total_movement);
    return 0;
}
```

Output:-

```
MINGW64/c/Users/STUDENTS/Desktop
STUDENT58M309-63 MINGW64 ~
$ cd ~/Desktop
STUDENT58M309-63 MINGW64 ~/Desktop
$ nano sstf.c
STUDENT58M309-63 MINGW64 ~/Desktop
$ gcc sstf.c -o sstf
STUDENT58M309-63 MINGW64 ~/Desktop
$ ./sstf
Enter number of requests: 8
Enter the request sequence:
98 183 41 122 14 124 65 67
Enter initial head position: 53
Total Head Movement = 323
STUDENT58M309-63 MINGW64 ~/Desktop
$
```

Output:-

```
MINGW64/c/Users/STUDENTS/Desktop
STUDENTSRM309-63 MINGW64 ~
$ cd ~/Desktop
STUDENTSRM309-63 MINGW64 ~/Desktop
$ nano sstf.c
STUDENTSRM309-63 MINGW64 ~/Desktop
$ gcc sstf.c -o sstf
STUDENTSRM309-63 MINGW64 ~/Desktop
$ ./sstf
Enter number of requests: 8
Enter the request sequence:
90 109 41 122 14 124 65 67
Enter initial head position: 53
Total Head Movement = 323
STUDENTSRM309-63 MINGW64 ~/Desktop
$
```